

Upgrade openSUSE to the Latest Linux Kernel

This article offers readers some interesting hacker tips. The tutorial is on upgrading the openSUSE 13.2 stock kernel to Linux Kernel 4.x. So go on and get your hands dirty!

OpenSUSE 13.2 ships with the stock Linux kernel 3.16.7-13. However, with all the interesting platform technologies for developers, like Docker and Limba, evolving at a rapid pace, there is a need for the functionality available in the later kernel versions.

The approach we adopt involves the following steps:

1. Use the command line tool *zypper*
2. Edit the */etc/zypp/zypp.conf* file to support multiple kernel configurations
3. Add the kernel HEAD repository
4. Use *zypper* to perform a distribution upgrade from the kernel HEAD repository
5. Reboot the system

First, let's edit the */etc/zypp/zypp.conf* file with the entries shown below.

The options are specified to support multiple kernels on the same system and retain multiple kernels including the oldest, i.e., the general availability (GA) kernel, the one openSUSE ships with. We also retain the running kernel.

```
multiversion = provides:multiversion(kernel)
multiversion.kernels = latest,latest-1,running,oldest
```

Second, let's add the openSUSE kernel HEAD repository that has the latest kernels. This applies irrespective of which kernel version is running on your system.

```
$ sudo zypper addrepo -f http://download.opensuse.org/
repositories/Kernel:/HEAD/standard/ kernel-repo
Adding repository 'kernel-repo' .....[done]
Repository 'kernel-repo' successfully added
```



```
Enabled      : Yes
Autorefresh  : Yes
GPG check    : Yes
URI          : http://download.opensuse.org/
repositories/Kernel:/HEAD/standard/
```

Next, perform a distribution upgrade only from the specific repository; in this case, the kernel HEAD repository.

```
$ sudo zypper dist-upgrade -r kernel-repo
```

```
Retrieving repository 'kernel-repo'
metadata
.....
.....[ ]
```

```
New repository or package signing key
received:
```

```
Repository:      kernel-repo
Key Name:        Kernel OBS
Project <kernel@build.opensuse.org>
Key Fingerprint: 4529410A B52F94C4
03B4B484 ECEE210 03579C1D
```

```
Key Created:     Tue 12 Feb 2013 05:52:10 PM IST
Key Expires:     Thu 23 Apr 2015 05:52:10 PM IST (expires
in 6 days)
Rpm Name:        gpg-pubkey-03579c1d-511a33f2
```

```
Do you want to reject the key, trust temporarily, or trust
always? [r/t/a/? shows all options] (r): a
Retrieving repository 'kernel-repo' metadata .....[done]
Building repository 'kernel-repo' cache .....[done]
Loading repository data...
Reading installed packages...
Computing distribution upgrade...
```

The following NEW package is going to be installed:

```
kernel-desktop-4.0-0-2.1.g38d8739
```

The following three packages are going to be upgraded:

```
kernel-docs kernel-firmware ucode-amd
```